Stereo Sum Embedding for Audio Steganography

Team 03

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July 10, 2025

Team Member Contributions

Ian Rohan- Did the introduction, and the functional block diagram
Quinn Westerberg- Contributed to the functional block diagram
Khristian Neal- Found the journal articles,and cited the sources

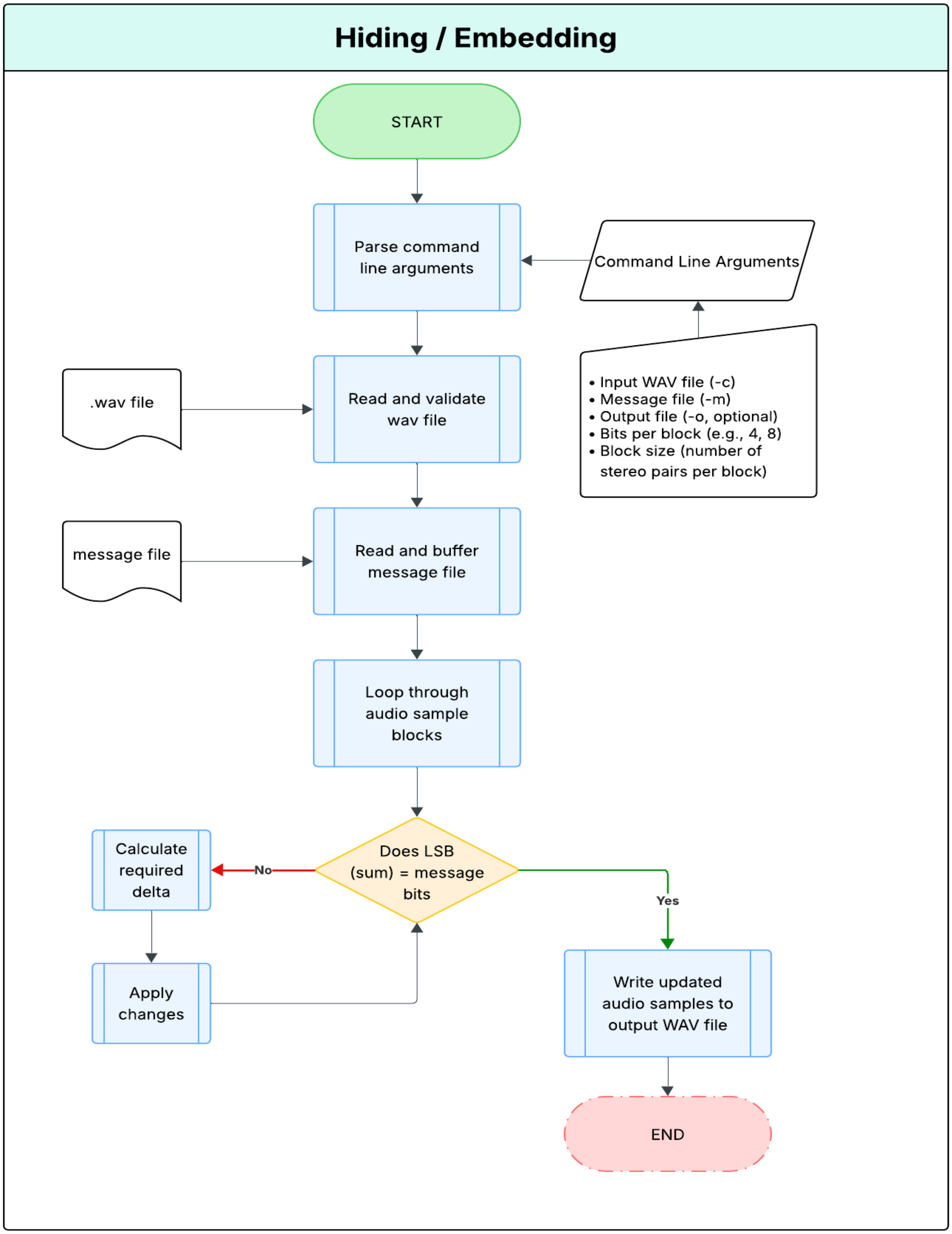
Introduction

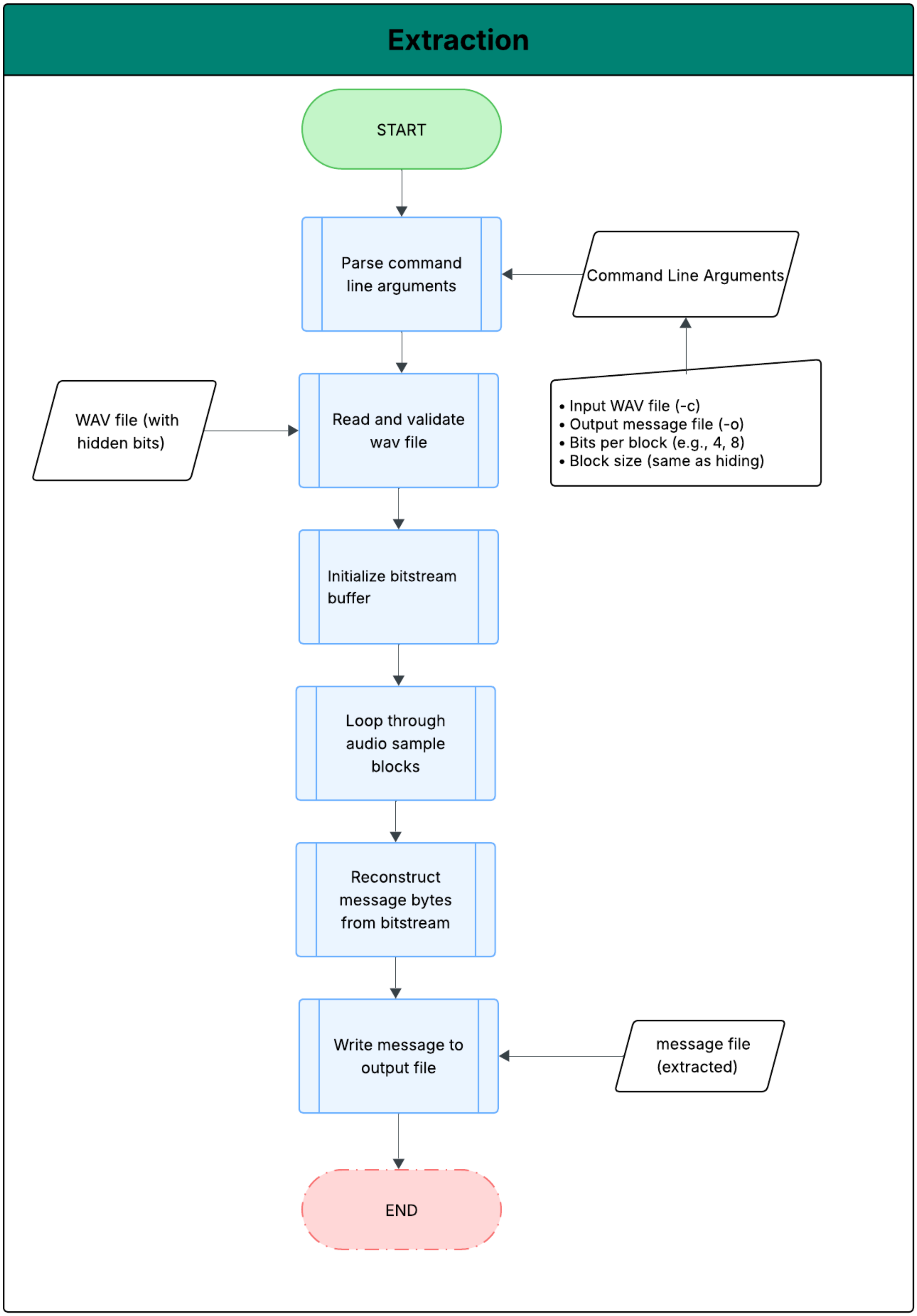
Our project focuses on developing a command-line steganographic tool for hiding and extracting secret data within 16-bit stereo WAV audio files using a custom sum-based embedding technique. Unlike traditional LSB methods, which simply overwrite the least significant bits of each sample and are easily detected, our approach embeds information in the least significant bits of the sum of multiple audio samples.

In particular, we consider a block of stereo audio samples (e.g., 4 or 8 samples), calculate their integer sum, and then modify the block—if necessary—so that the least significant bits of the sum equal the desired part of the message. This permits us to embed a variable number of message bits per block (e.g., 1–8), allowing a tradeoff between capacity and imperceptibility. If the sum already contains the correct bits, the block is not altered. Otherwise, the program determines the minimal possible modifications to one or more of the samples in the block to make the sum agree, while keeping the changed values within the legal 16-bit signed interval (−32,768 to 32,767).

The end result will be a tool that enables users to embed a message file into an uncompressed audio file with user-defined parameters like number of bits per block. The extraction mode will complement this by doing the reverse: reading back the embedded data from the sum of the same-sized sample blocks. This project will try to achieve equilibrium between performance, audio quality, and security through a well-thought-out and instructor-approved steganographic technique.

Functional Block Diagram




Extraction
START
Parse command line arguments
Command Line Arguments
• Input WAV file (-c)
• Output message file (-o)
• Bits per block (e.g., 4, 8)
• Block size (same as hiding)
WAV file (with hidden bits)
Read and validate wav file
Initialize bitstream buffer
Loop through audio sample blocks
Reconstruct message bytes from bitstream
Write message to output file
message file (extracted)
END

Works Cited

Hwang, Min-Jae, et al. "SVD-Based Adaptive QIM Watermarking on Stereo Audio Signals." *IEEE Xplore,* 29 June 2017. https://ieeexplore-ieee-org.libweb.lib.utsa.edu/document/7962215. Accessed 10 July 2025.

Schuijers, Erik, et al. "Parametric Coding of Stereo Audio." *SpringerOpen, V*ol. 2005, no. 9, 21 June 2005. https://asp-eurasipjournals.springeropen.com/articles/10.1155/ASP.2005.1305, Accessed 10 July 2025.